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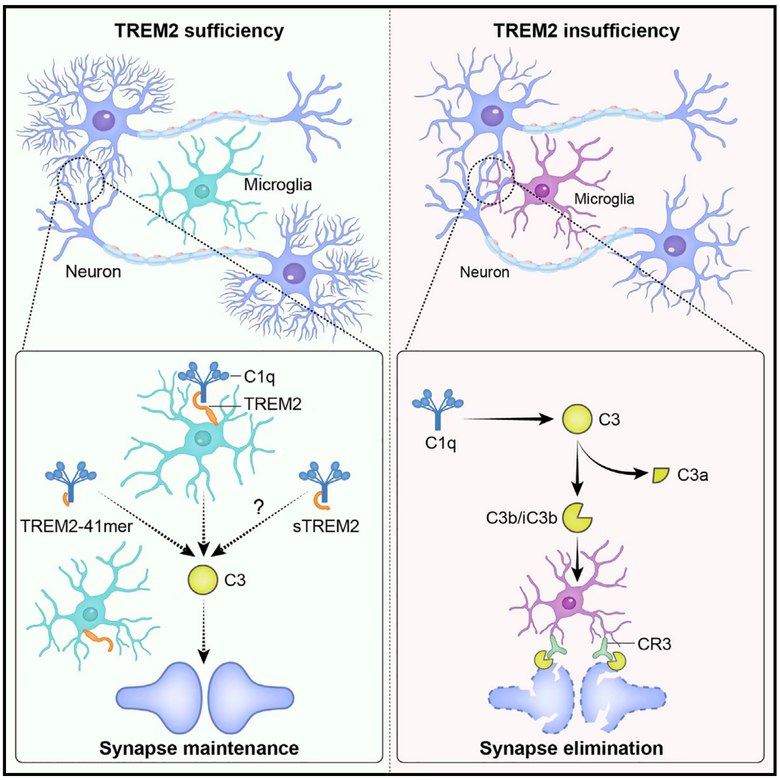
TREM2 receptor protects against complement-mediated synaptic loss by binding to complement C1q during neurodegeneration

Graphical abstract



Highlights

- TREM2 interacts with C1q *in vitro* and *in vivo*
- TREM2 suppresses the classical complement cascade by binding to C1q
- *Trem2* insufficiency increases complement-mediated synaptic loss in AD mice models
- A TREM2 peptide that binds to C1q rescues synaptic impairments in AD mouse models

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In brief

TREM2 is strongly linked to Alzheimer's disease risk, but its functions are not fully understood. Zhong et al. discovered that TREM2 suppresses complement activity and synaptic loss by binding to C1q. They identified a 41-amino-acid TREM2 peptide that restricts complement-mediated synaptic elimination during neurodegeneration.

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TREM2 receptor protects against complement-mediated synaptic loss by binding to complement C1q during neurodegeneration

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SUMMARY

Triggering receptor expressed on myeloid cells 2 (TREM2) is strongly linked to Alzheimer's disease (AD) risk, but its functions are not fully understood. Here, we found that TREM2 specifically attenuated the activation of classical complement cascade via high-affinity binding to its initiator C1q. In the human AD brains, the formation of TREM2-C1q complexes was detected, and the increased density of the complexes was associated with lower deposition of C3 but higher amounts of synaptic proteins. In mice expressing mutant human tau, *Trem2* haploinsufficiency increased complement-mediated microglial engulfment of synapses and accelerated synaptic loss. Administration of a 41-amino-acid TREM2 peptide, which we identified to be responsible for TREM2 binding to C1q, rescued synaptic impairments in AD mouse models. We thus demonstrate a critical role for microglial TREM2 in restricting complement-mediated synaptic elimination during neurodegeneration, providing mechanistic insights into the protective roles of TREM2 against AD pathogenesis.

INTRODUCTION

Synaptic loss is an early and invariant feature of Alzheimer's disease (AD), which strongly correlates with the severity of cognitive dysfunction. Recent studies in animal models of AD have suggested that microglia and the complement-dependent pathway that prune excess synapses during development are aberrantly activated, contributing to synaptic loss in AD.[1] As initiators of these events, both aggregated amyloid-β (Aβ) and tau have been demonstrated to activate the classical complement pathway in biochemical assays.[2–4] The first component of the classical pathway, C1q, is highly upregulated and accumulates at synapses in both the amyloidosis and tauopathy mouse models of AD, suggesting an induction of the complement pathway by AD pathological proteins *in vivo*.[5,6] Upon activation, the C1q triggers a protease cascade leading to the deposition of the downstream complement protein C3. The C3 cleavage products (C3b and iC3b) can directly activate the complement receptor 3 (CR3) on microglia, leading to synaptic elimination via phagocytosis.[7] Genetic deletion of C1q, C3, or CR3 reduces the number of phagocytic microglia and the extent of synaptic loss in mouse models of amyloidosis.[6] Similarly, blocking complement function by deleting C3 ameliorates synaptic loss and neurodegeneration in the tauopathy mouse model.[8] The deficiency of C3 protects against plaque-related synaptic loss and cognitive decline in aged APP/PS1 mice despite enhancing plaque burden.[9] Inactivation of the complement receptors on microglia for C3a (C3aR1) or C5a (C5aR1) also attenuates neuroinflammation, synaptic deficits, and neurodegeneration in mouse models of AD.[10,11] These data suggest that targeting key components in the complement activation cascade could improve disease outcomes in the presence of AD pathologies. Indeed, several anti-complement therapeutic strategies have been proposed and tested in a variety of preclinical AD models. The C5aR antagonist peptide (PMX205) decreases pathology and enhances behavioral performance in two mouse models of AD.[12] Additionally, a C1q-blocking antibody also prevents complement-mediated synapse removal by microglia and rescues synapse density in the P301S tauopathy mouse model.[5]

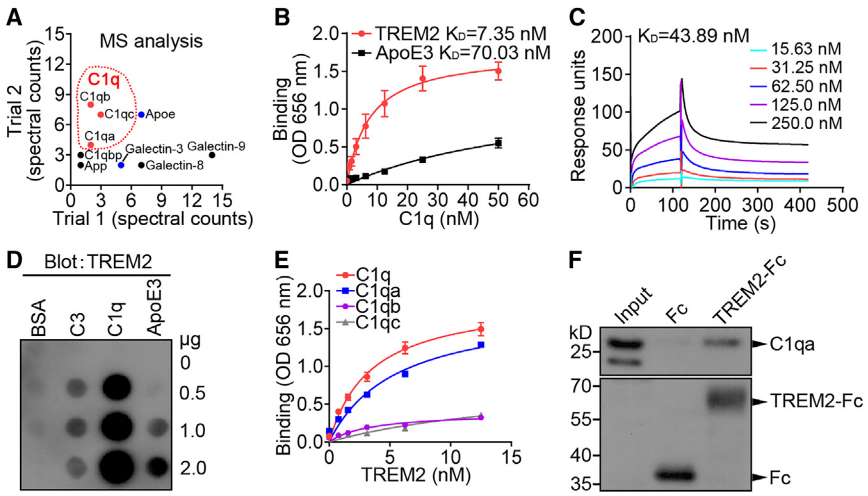


Figure 1. TREM2 binds directly to C1q

(A) Spectral counts of proteins identified by MS from two independent IP experiments.

(B) Binding of C1q to immobilized TREM2 (purified TREM2 1–171 aa) or ApoE3 was determined by solid-phase binding (SPB) assay (three independent experiments; n = 6).

(C) The binding profile of TREM2 to different concentrations of C1q was measured by surface plasmon resonance assay.

(D) Representative dot blot analysis of TREM2 binding to increasing amounts of purified C3, C1q, and apoE3. BSA served as a negative control.

(E) Binding of TREM2 to immobilized C1q, C1qa, C1qb, or C1qc was determined by SPB assay (three independent experiments; n = 6).

(F) The Fc or TREM2 (1–171 aa)-Fc was pre-bound to protein A agarose beads and used as baits to pull down C1qa from the brain lysates of *Trem2*-KO mice. The precipitated products were analyzed by western blotting.

All data are represented as mean ± SEM.

See also Figure S1.

The complement system modulates tissue homeostasis and interacts with the innate and adaptive immune system to support immune surveillance. A cascade of complement regulators exists to regulate a proportionate balance of the complement activity. To prevent runaway activation and damage to self-tissue, the complement system is kept under tight control by a variety of complement inhibitors in the peripheral immune system.[13] However, the possibility of harnessing complement activation by endogenous inhibitors is only just beginning to be explored in the central nervous system. One such example is the apolipoprotein E (apoE), which inhibits the classical complement cascade (CCC) by binding to the C1q with high affinity.[14] The C1q-apoE complex forms an active disease-relevant regulatory module in AD and atherosclerosis by attenuating unresolvable inflammation. Another example is the sushi repeat protein X-linked 2 (SRPX2), which directly binds to C1q and inhibits complement activation, leading to the protection against complement-mediated synaptic elimination during development.[15]

The triggering receptor expressed on myeloid cells 2 (TREM2) is a single-transmembrane immune receptor exclusively expressed in microglia within the central nervous system.[16] *TREM2* is recognized as the strongest immune-specific genetic risk factor for AD because its heterozygous R47H variant confers a 3-to-4-fold increased risk of AD.[17] The extracellular immunoglobulin-like domain of TREM2 binds to an array of pathological molecules in AD, including apoE and Aβ.[18] The engagement of ligand triggers several signaling events downstream of TREM2, impacting an array of microglial functions, including inflammation, phagocytosis, proliferation, migration, survival, and metabolism.[16,19] TREM2 is also essential for microglia-mediated synaptic pruning and normal brain connectivity during the early developmental stage as well as adulthood.[20,21] Given the possible involvement of excessive synaptic elimination in AD pathogenesis, we investigated whether TREM2 regulates microglia-mediated synaptic pruning in the context of AD.

Here, we provide evidence that TREM2 serves as a complement inhibitor that protects synapses against complement-mediated elimination in the brains of AD mouse models and AD patients. We show that the extracellular domain of TREM2 directly binds to C1q, the initiator of the classical complement pathway, and restrains complement activation *in vitro*. By means of a proximity ligation assay (PLA), the formation of TREM2-C1q complexes was detected in the human AD brain where its density was inversely correlated to the amounts of C3 deposition. In mice expressing mutant human tau (PS19), *Trem2* haploinsufficiency increased complement-mediated microglial engulfment of synapses and accelerated synaptic loss. Importantly, we identified a 41-amino-acid sequence in TREM2 (41mer) that preserves the binding affinity to C1q and prohibits complement activation by *in vitro* biochemical assays. Administration of the 41mer peptide reduced complement activation, rescued synaptic deficits and neurodegeneration in both the amyloidosis and tauopathy mouse models of AD. Our findings implicate that microglial TREM2 regulates complement-mediated synaptic elimination in neurodegeneration.

RESULTS

TREM2 binds directly to C1q

To gain insights into the molecular mechanisms underlying TREM2 function, we performed affinity-purification coupled to mass spectrometry with the TREM2-Fc fusion construct and found that TREM2-Fc, but not the Fc alone, precipitated known TREM2 ligands from the brain lysates of *Trem2*-knockout (KO) mice, including apoE and galectin-3, as previously reported (Figures 1A and S1A).[22–25] These unbiased pull-down experiments revealed new binding partners for TREM2, including C1qa, C1qb, and C1qc, which are the three components of C1q. The binding of TREM2 to C1q was further confirmed by a solid-phase binding assay (Figure 1B). It is noteworthy that TREM2 binds to C1q with even higher affinity than apoE, which has been shown to attenuate the CCC activity via complex formation with C1q.[14] The surface plasmon resonance assay and dot blot binding assay also showed a dose-dependent binding of

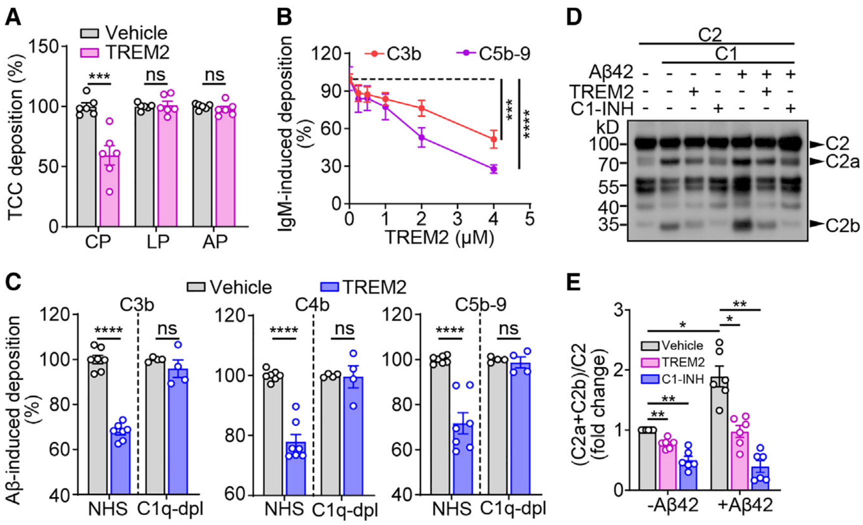


Figure 2. TREM2 inhibits C1q-initiated classical complement activation

(A) TREM2 specifically inhibits the classical complement pathway. TREM2 protein (2 μM) in NHS was added to microtiter plates coated with IgM, mannan, or zymosan. The TCC deposition was determined using C5b-9 antibody (three independent experiments; n = 6 per group, unpaired Student's t test). CP: classical pathway; LP: lectin pathway; AP: alternative pathway.

(B) Increasing amounts of purified TREM2 protein in NHS were added to IgM-coated microtiter plates, then C3b or C5b-9 deposition was analyzed by specific antibodies (two independent experiments; n = 4 per group, one-way ANOVA).

(C) TREM2 protein (2 μM) in NHS or C1q-depleted human serum (C1q-dpl) was added to fAβ42-coated microtiter plates, then C3b, C4b, and C5b-9 deposition were determined using C3, C4, and C5b-9 antibodies, respectively (three independent experiments for the NHS group, two independent experiments for the C1q-dpl group; n = 7 for the NHS group, n = 4 for the C1q-dpl group, unpaired Student's t test).

(D) TREM2 recombinant protein (4 μM) or C1-INH was incubated with C2 in the presence of C1 supplemented with or without fAβ42. C2 cleavage was analyzed by western blotting.

(E) Quantitation of the protein amounts of (C2a + C2b)/C2 in (D) (six independent experiments, n = 6 per group, two-way ANOVA).

All data are represented as mean ± SEM. *p < 0.05; **p < 0.01; ***p < 0.001; ****p < 0.0001; ns, not significant.

See also Figure S1.

C1q to TREM2 recombinant protein (Figures 1C and 1D). Although AD-associated TREM2 variants have been shown to impair the binding of TREM2 to its ligands,[26,27] we found that the R47H and D87N mutants bound to C1q with comparable affinity to the wild-type TREM2 protein (Figure S1B). TREM2 binding to complement proteins revealed strong and specific binding to C1q and C1, but not to C2 and C3 (Figure S1C). The solid-phase binding assay revealed a specific interaction between TREM2 and the C1qa subunit of C1q, whereas neither the C1qb nor C1qc bound to TREM2 (Figure 1E). The endogenous C1qa from the brain lysates of *Trem2*-KO mice also bound to TREM2 immobilized on beads, consistent with the binding of recombinant C1qa to TREM2 (Figure 1F). The residues encompassing 14–26 of the C1qa polypeptide chain have been shown to mediate the binding of C1q to the antibody-independent complement activators.[3,28,29] The peptides containing residues 14–26 of C1qa significantly blocked C1q binding to TREM2, implicating that the 14–26 region of C1qa plays a role in mediating C1q-TREM2 interaction (Figure S1D). Taken together, these experiments indicate that TREM2 directly binds to C1q via recognizing its C1qa subunit.

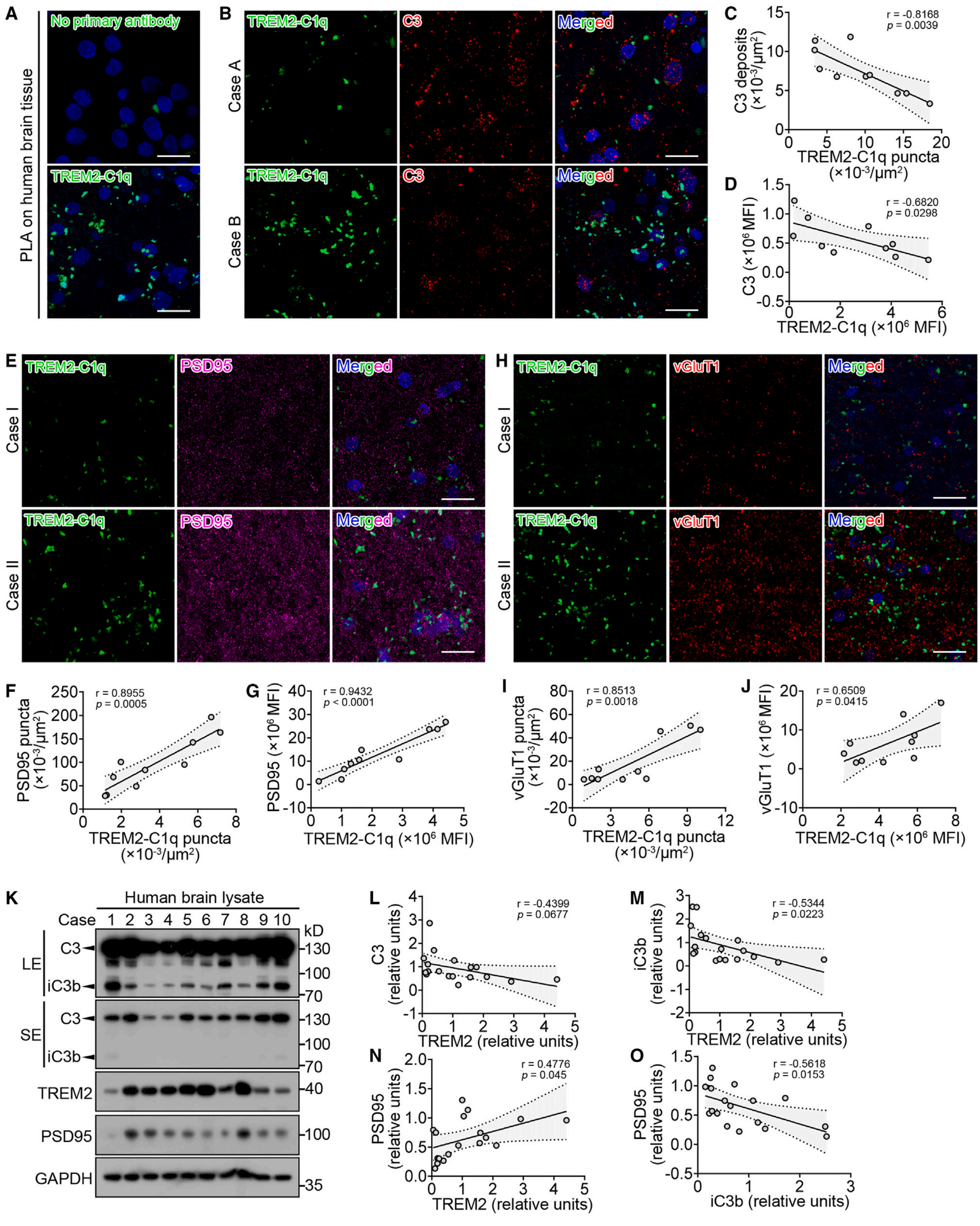
TREM2 suppresses the classical complement cascade

Next, we sought to determine the functional effect of TREM2 binding to C1q. TREM2 reporter cells (TD4), which express both TREM2 and DAP12 and produce β-galactosidase upon TREM2 engagement,[30] were utilized to determine if C1q can initiate TREM2 downstream signaling. As a positive control, the oligomeric Aβ42 (oAβ42) stimulated the expression of β-galactosidase in TREM2 reporter cells (Figure S1E). In contrast, the presence of C1q failed to stimulate the reporter activity in TD4 cells, thus excluding the possibility that C1q acts as a canonical TREM2 ligand.

Because C1q is the first component of the classical complement pathway, next we tested the roles of TREM2 in the complement activation. In a complement-mediated killing assay, *Escherichia coli* were killed in the presence of normal human serum (NHS). However, *Escherichia coli* remained viable when the NHS was supplemented with TREM2-Fc, but not the Fc alone control, indicating a role of TREM2 in suppressing the complement activation pathway (Figures S1F and S1G). Furthermore, TREM2 inhibited the deposition of the terminal complement complex (TCC, C5b-9) when the complement was activated via the classical but not the alternative or lectin pathway, indicating that TREM2 specifically inhibits the CCC activity, but not the alternative or lectin pathway (Figure 2A). TREM2 also suppressed the deposition of C3b and C5b-9 in a dose-dependent manner (Figure 2B). Fibrillary Aβ (fAβ) has been shown to bind to C1q, resulting in the activation of the CCC pathway.[3,31] We found that purified TREM2 inhibited the CCC and reduced the deposition of C3b, C4b, and C5b-9 when the CCC was activated by fAβ (Figure 2C). Notably, TREM2 failed to inhibit the CCC when C1q was depleted from the human serum, indicating that TREM2 inhibits the CCC activity in a C1q-dependent manner. During CCC initiation, C1q undergoes a conformational change, allowing the binding of proteases C1s and C1r to form the C1 complex, which further cleaves the C2 or C4.[32] To determine whether TREM2 inhibits CCC activation at the C1 step, we further tested the ability of TREM2 to reduce C2 cleavage by C1 *in vitro*. We found that the presence of purified TREM2 decreased the cleavage of C2 by C1 in both the presence and absence of fAβ, suggesting that TREM2 blocks CCC activation at the C1 step (Figures 2D and 2E). Taken together, our data revealed that TREM2 acts as a specific CCC inhibitor by high-affinity binding to C1q.

TREM2 and C1q form complexes in the human AD brain

We further validated the interaction between TREM2 and C1q in physiological context by means of a PLA. PLA is a sensitive and



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specific technique to visualize protein-protein interaction occurring within 40 nm at endogenous protein expression levels.[33] In the hippocampi of the post-mortem human AD brain, PLA signals corresponding to TREM2-C1q interaction were observed (Figure 3A). Inhibition of the classical complement pathway prevents C3 deposition and synaptic loss in the AD brain.[6] Based on our *in vitro* observation that TREM2 blocks CCC activation, we further determined the functional relevance of TREM2 interaction with C1q *in vivo* by analyzing the correlation between the density of TREM2-C1q PLA signals and those of C3 and PSD95. Higher density of TREM2-C1q PLA signals tended to be associated with fewer C3 deposition (Figure 3B). In addition, the number and mean fluorescence intensity (MFI) of TREM2-C1q puncta were significantly and inversely correlated with those of C3 deposits (Figures 3C and 3D). In contrast, the amount and density of TREM2-C1q puncta were positively correlated with those of PSD95 (Figures 3E–3G) or vesicular glutamate transporter 1 (vGluT1) foci (Figures 3H–3J). We further examined the protein amounts of TREM2, C3, and PSD95 in post-mortem AD brains (Figure 3K). An inverse correlation was found between the protein amounts of TREM2 and C3/iC3b (Figures 3L and 3M), as well as between the protein amounts of PSD95 and iC3b (Figure 3O). However, we observed a significant positive correlation between the protein amounts of TREM2 and PSD95 (Figure 3N). Taken together, our PLA data suggest that the TREM2-C1q complexes form in human AD brains. The presence of TREM2-C1q complexes positively correlates with the amounts of synaptic proteins, whereas negatively correlating with C3 deposition. These data indicate that the formation of TREM2-C1q complex may play a protective role in the pathogenesis of AD.

Trem2 haploinsufficiency increases microglial engulfment of synapses and accelerates synaptic loss in PS19 mice

Next, we investigated whether the loss of *Trem2 in vivo* leads to complement activation in the brains of AD mouse models. The hemizygous TREM2 variants are associated with elevated risk for AD, whereas the complete loss of TREM2 in humans results in Nasu-Hakola disease.[34] Therefore, we chose to use the mice with *Trem2* haploinsufficiency, which may be a more relevant model to study how TREM2 contributes to AD pathogenesis. In the 5xFAD mice, the amounts of C3 and C1qa were significantly reduced upon *Trem2* haploinsufficiency (Figures S2A and S2B). However, compared with the 5xFAD/*Trem2*$^{+/+}$ mice, the 5xFAD/*Trem2*$^{+/-}$ mice also had reduced Iba1 reactivity and the number of microglia, which is consistent with a previous report (Figures S2C–S2E).[35] Because microglia are the dominant source of C1qa in the brain (Figure S2F),[36] the decreased amounts of C3 and C1qa in the 5xFAD/*Trem2*$^{+/-}$ mice might be attributed to less microglia than the 5xFAD/*Trem2*$^{+/+}$ mice. Therefore, the 5xFAD mouse model is not suitable for addressing the effects of *Trem2* loss on complement activation.

The PS19 mice exhibit significant tau inclusions, microgliosis, and neuronal loss at 8–9 months of age.[37] In contrast to the 5xFAD mice, *Trem2* haploinsufficiency did not induce significant changes in Iba1 reactivity and the number of microglia in the 6-month-old PS19 mice (Figures S2G–S2I), consistent with a previous study examining *Trem2*$^{+/+}$ and *Trem2*$^{+/-}$ microglia in the 9-month-old PS19 mice.[38] Although *Trem2* haploinsufficiency has been reported to exacerbate tau pathology in the 9-month-old PS19 mice,[38] no such changes were observed in the 6-month-old PS19 mice (Figures S2J and S2K). After excluding the unwanted effects on microgliosis and tau pathology, we next assessed whether *Trem2* haploinsufficiency leads to complement activation in the 6-month-old PS19 mice.

Equivalent protein amounts of C1qa were observed in the brain of *Trem2*$^{+/+}$ and *Trem2*$^{+/-}$ PS19 mice, but significant increases in the amounts of C3 and iC3b protein were observed in the brains of PS19/*Trem2*$^{+/-}$ mice (Figures 4A and 4B), suggesting a role of Trem2 in blocking complement activation at the C1 step. In contrast, no significant changes in the protein

Figure 3. TREM2-C1q complexes correlate with complement activity and the amounts of synaptic protein in the human AD brain

(A) Human brain sections were co-stained for TREM2 and C1q, and the interaction between TREM2 and C1q was detected by *in situ* PLA. Negative control represents the background staining observed in the PLA reaction in the absence of primary antibodies. Green for TREM2-C1q complexes and blue for nuclei. Scale bars, 20 μm.

(B) Human AD brain hippocampal sections were examined by PLA for the presence of TREM2-C1q complexes (green) and co-stained with C3 antibody (red). Scale bars, 20 μm.

(C) Correlation between the number of TREM2-C1q complexes puncta and the number of C3 deposits in the human AD brain (four independent experiments; n = 10).

(D) Correlation between the MFI of TREM2-C1q complexes and the MFI of C3 in the human AD brain (four independent experiments; n = 10).

(E) Human AD brain hippocampal sections were examined by PLA for the presence of TREM2-C1q complexes (green) and co-stained with PSD95 antibody (magenta). Scale bars, 20 μm.

(F) Correlation between the number of TREM2-C1q complexes puncta and the number of PSD95 puncta in the human AD brain (three independent experiments; n = 10).

(G) Correlation between the MFI of TREM2-C1q complexes and the MFI of PSD95 in the human AD brain (three independent experiments; n = 10).

(H) Human AD brain hippocampal sections were examined by PLA for the presence of TREM2-C1q complexes (green) and co-stained with vGluT1 antibody (red). Scale bars, 20 μm.

(I) Correlation between the number of TREM2-C1q complexes puncta and the number of vGluT1 puncta in the human AD brain (two independent experiments; n = 10).

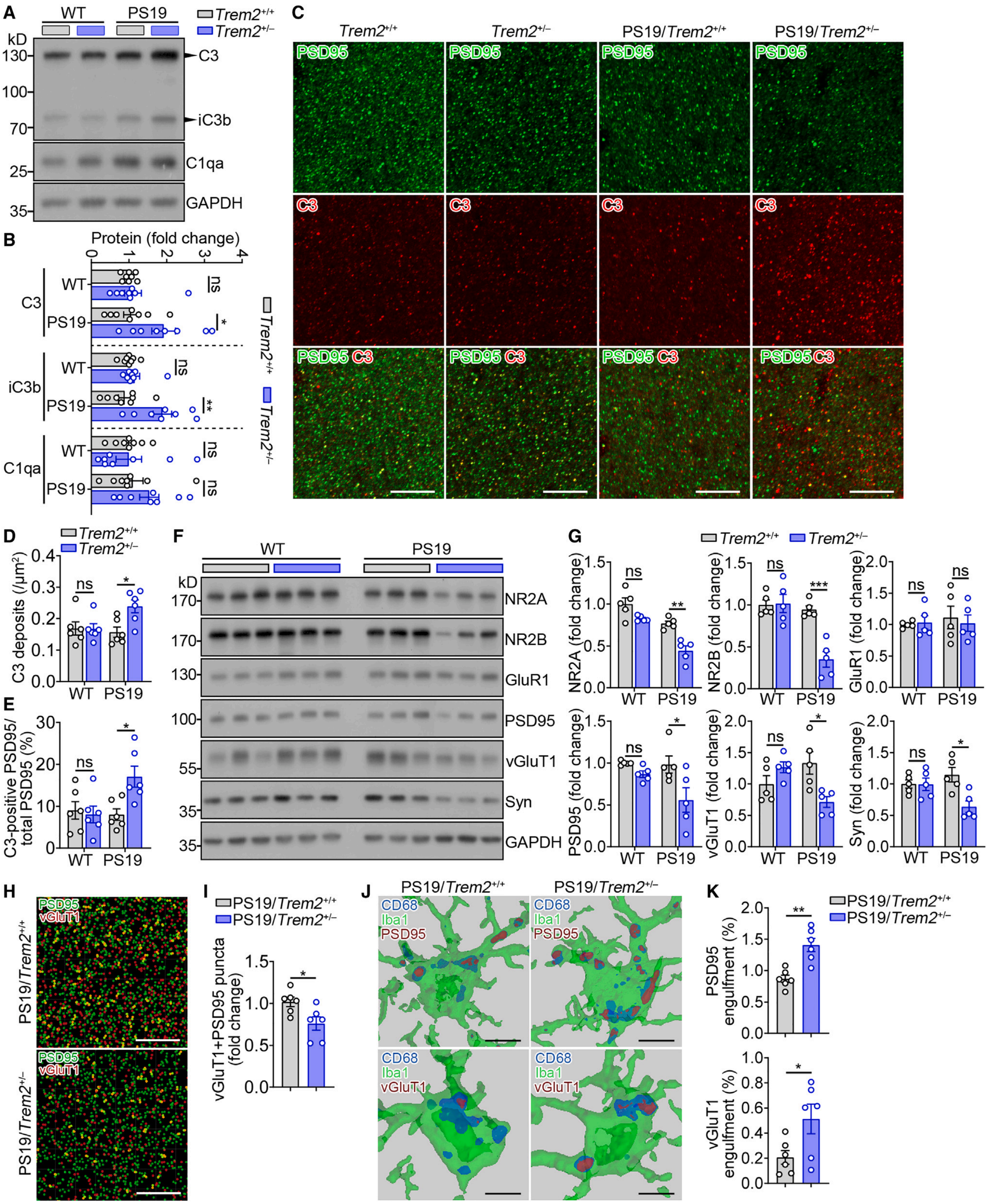
(J) Correlation between the MFI of TREM2-C1q complexes and the MFI of vGluT1 in the human AD brain (two independent experiments; n = 10).

(K) Representative immunoblots of cortex samples from AD patients probed with TREM2, C3, and PSD95 antibodies. LE, long exposure; SE, short exposure.

(L–O) Correlation between the protein amounts of TREM2 and C3 (L), TREM2 and iC3b (M), TREM2 and PSD95 (N), and iC3b and PSD95 (O) (four independent experiments; n = 18).

r represents the correlation coefficient; p is significance.

See also Table S1.



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amounts of C1qa, C3, or iC3b were observed in the brain of non-transgenic (WT) *Trem2*$^{+/+}$ and *Trem2*$^{+/-}$ mice, indicating a specific blocking of complement activation by Trem2 in the disease context. We further performed immunofluorescence studies on C3 and PSD95 and found increased C3 deposition in the brains of PS19/*Trem2*$^{+/-}$ mice (Figures 4C and 4D). A higher percentage of C3 was found to colocalize with PSD95 in the hippocampi of PS19/*Trem2*$^{+/-}$ mice than in that of PS19/*Trem2*$^{+/+}$ mice (Figure 4E), suggesting that more synapses may be marked for elimination upon partial *Trem2* deficiency. To better understand synaptic alterations upon *Trem2* deficiency, we examined the amounts of various proteins involved in synaptic plasticity and function. The amounts of presynaptic vGluT1 and synaptophysin (Syn), postsynaptic N-methyl-D-aspartate (NMDA) receptor subunits (NR2A and NR2B), and PSD95 were significantly decreased in the PS19/*Trem2*$^{+/-}$ mice compared with PS19/*Trem2*$^{+/+}$ mice (Figures 4F and 4G). Quantification of colocalized pre- and post-synaptic puncta (vGluT1^{+} and PSD95^{+}) revealed a significant loss of synapses upon *Trem2* deficiency in PS19 mice (Figures 4H and 4I). To investigate the role of microglia in the synaptic density reduction observed in PS19/*Trem2*$^{+/-}$ mice, we measured microglial engulfment of synaptic elements by quantitating the amount of PSD95 or vGluT1 within microglial phagolysosomes. Significantly more PSD95 and vGluT1 puncta were found in CD68^{+} microglial structures in the PS19/*Trem2*$^{+/-}$ mice compared with PS19/*Trem2*$^{+/+}$ mice (Figures 4J and 4K). These data together suggest that microglial *Trem2* deficiency decreases synaptic density by increasing complement-mediated microglial engulfment of synapses in the tauopathy mouse model.

TREM2 binds to C1q and inhibits the classical complement pathway via residues 31–71

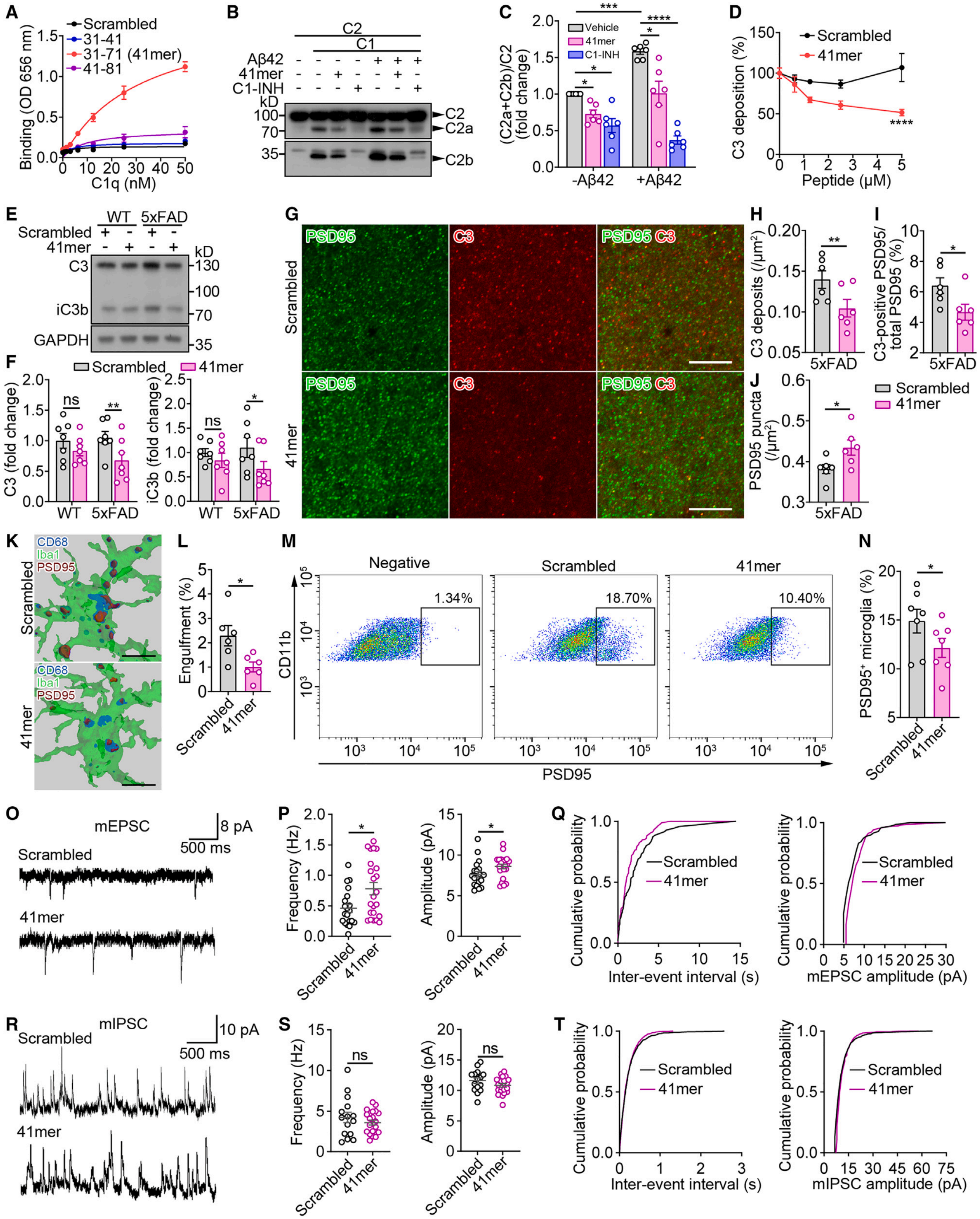
To experimentally determine which regions of TREM2 contribute to C1q binding, we performed serial truncations on the extracellular domain of human TREM2 (1–171) (Figures S3A and S3B). In the C1q binding assay, mutants carrying residues 1–31 showed no detectable binding to C1q, whereas the binding affinity of the 1–71 fragment to C1q was similar to that of the 1–171 fragment (Figures S3C and S3D). These data suggest that the residues 31–71 of TREM2 may be necessary for its interaction with C1q. In the solid-phase binding assay, the synthetic TREM2 peptide (31–71, 41mer) bound to C1q in a dose-dependent manner (Figure 5A). Moreover, the biotin-labeled 41mer peptide precipitated C1qa but not apoE or App from the brain lysates of *Trem2*-KO mice, suggesting a specific interaction between the TREM2 31–71 peptide and C1q (Figure S3E). Consistent with our previous reports, the TREM2 41–81 bound to oAβ and increased the expression of proinflammatory cytokines in microglia (Figures S3F and S3G).[39] In contrast, the TREM2 31–71 peptide barely bound to oAβ and had no significant impacts on microglial inflammation. We further observed that neither the TREM2 41–81 nor 31–41 peptide bound to C1q with measurable levels (Figure 5A). Our results hence emphasized the specific roles of TREM2 residues 31–71 and 41–81 in the binding of C1q and oAβ, respectively. Next, we sought to determine the functional effects of TREM2 31–71 peptide binding to C1q. Similar to the full-length extracellular domain of TREM2, preincubation of the TREM2 31–71 peptide with purified C1 decreased its ability to cleave C2 in both the presence and absence of fAβ (Figures 5B and 5C). Furthermore, TREM2 31–71 peptide reduced the deposition of C3 when the CCC was activated by IgM (Figure 5D). Taken together, these data indicate that the TREM2 31–71 peptide is necessary and sufficient to bind to C1q and inhibit the classical complement pathway *in vitro*.

A synthetic peptide encompassing the residues 31–71 of TREM2 alleviates microglial engulfment of synapses and ameliorates synaptic plasticity in 5xFAD mice

To determine whether the synthetic TREM2 31–71 peptide inhibits complement activation *in vivo*, we injected the TREM2 peptide and the scrambled peptide control into the right and left hippocampi of 7-month-old WT or 5xFAD amyloid model mice, respectively. 7 days after delivery, the TREM2 peptide significantly reduced the amounts of C3 and iC3b protein in 5xFAD mice but had no significant effect in WT mice (Figures 5E and 5F). Of note, the TREM2 41–81 peptide, which does not bind to C1q, had no impact on the protein amounts of C3 or iC3b in 5xFAD mice (Figures S3H and S3I), suggesting that the interaction with C1q is a prerequisite for the TREM2 peptide to reduce C3 deposition. Immunofluorescence studies also revealed a reduction of C3 deposition and a lower percentage of

Figure 4. TREM2 haploinsufficiency exacerbates complement activation and complement-mediated synaptic loss in PS19 mice

(A) Representative western blots of indicated proteins in hippocampal lysate from 6-month-old *Trem2*$^{+/+}$, *Trem2*$^{+/-}$, PS19/*Trem2*$^{+/+}$, and PS19/*Trem2*$^{+/-}$ mice.
(B) Quantitation of C3, iC3b, and C1qa amounts in (A) (four independent experiments; n = 8 mice per group, two-way ANOVA).
(C) Representative images from the hippocampal CA1 region of these four mouse groups stained for PSD95 (green) and C3 (red). Scale bars, 10 μm.
(D and E) Quantitation of the density of C3 deposits (D) and the percentage of C3 colocalized with PSD95 (E) in (C). Data points represent the average of three technical repeats (one experiment; n = 6 mice per group, two-way ANOVA).
(F) Synaptic proteins from the hippocampi of four mouse groups were analyzed by western blotting. Representative immunoblots of three mice per group are shown.
(G) Quantitation analysis of protein amounts in (F) (two independent experiments; n = 5 mice per group, two-way ANOVA).
(H) Three-dimensional rendering of confocal stacks of PSD95 (green) and vGluT1 (red) puncta in the hippocampal CA1 region of PS19/*Trem2*$^{+/+}$ and PS19/*Trem2*$^{+/-}$ mice at 6 months of age. Scale bars, 10 μm.
(I) Quantitation of the density of the PSD95 signal that overlaps with vGluT1 puncta. Data points represent the average of three technical repeats (one experiment; n = 6 mice per group, unpaired Student's t test).
(J) Three-dimensional rendering of confocal stacks of microglia and engulfed PSD95 or vGluT1 puncta (detected within microglial CD68^{+} structures) of PS19/*Trem2*$^{+/+}$ and PS19/*Trem2*$^{+/-}$ mice. Scale bars, 5 μm.
(K) Quantitation of PSD95 or vGluT1 engulfment within CD68^{+} microglial structures. Data points represent the average of three technical repeats (one experiment; n = 6 mice per group, unpaired Student's t test).
All data are represented as mean ± SEM. *p < 0.05; **p < 0.01; ***p < 0.001; ns, not significant.
See also Figure S2.



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C3 colocalized with PSD95 in 5xFAD mice after being administered with the TREM2 31–71 peptide (Figures 5G–5I). Furthermore, the 41mer peptide prevented microglial engulfment of synapse when the ratio of PSD95$^+$ (Figures 5K–5N) or vGluT1$^+$ (Figures S5I and S5J) puncta in microglia was quantitated by either immunostaining or flow cytometric analysis. To better understand synaptic alterations upon TREM2 peptide administration, we examined synaptic number and found higher PSD95$^+$ puncta intensity than control (Figure 5J). Furthermore, we demonstrated that the TREM2 peptide increased the frequency and amplitude of miniature excitatory postsynaptic currents (mEPSCs) in CA1 neurons of 5xFAD mice (Figures 5O–5Q). The amplitude and frequency of miniature inhibitory postsynaptic currents (mIPSCs) did not significantly differ between the TREM2 peptide and the scrambled peptide-treated groups (Figures 5R–5T).

To further investigate whether the TREM2 31–71 peptide has direct impacts on neurons, primary neuronal cultures either cultured alone or co-cultured with microglia were treated with the TREM2 31–71 peptide. The TREM2 31–71 peptide did not affect the synaptic density or the abundance of synaptic proteins in the pure primary cultures of neurons (Figures S3L–S3O). Consistently, the amplitude and frequency of both mEPSCs and mIPSCs did not significantly differ between neurons treated with the TREM2 31–71 peptide or the scrambled peptide (Figures S3P–S3S). In contrast, in the neuron-microglia co-cultures, the TREM2 31–71 peptide significantly increased the synaptic density and the abundance of synaptic proteins compared with the scrambled peptide control (Figures S3L–S3O). These data hence suggest that the TREM2 31–71 peptide does not directly affect neuronal properties.

We previously reported that the soluble form of TREM2 enhances synaptic integrity and plasticity through restoring beneficial microglial activity and reducing plaque pathology.[40] However, no changes in the Aβ burden or microglial responses were observed upon TREM2 peptide administration (Figures S4A–S4F), suggesting that the TREM2 peptide may directly modulate synaptic function by preventing microglial engulfment of synapses. In the 12-month-old 5xFAD mice, which represent late-stage disease progression, the TREM2 peptide also appeared to restrain the pathologic complement activation and the complement-mediated microglial engulfment of synapses (Figures S5A–S5H). These data together suggest that the TREM2 31–71 peptide enhances synaptic density and plasticity by reducing complement-mediated microglial engulfment of synapses in the mouse model of amyloidosis.

The synthetic TREM2 peptide suppresses complement-mediated synaptic loss and enhances cognitive function in PS19 mice

Next, we sought to determine whether the synthetic TREM2 41mer peptide inhibits complement activation in the tauopathy mouse model. 7 days after stereotaxic injection, the TREM2 31–71 peptide significantly reduced the amounts of C3 and iC3b protein in the hippocampi of 12-month-old PS19 mice (Figures 6A–6C), indicating suppression of the complement activation. In contrast, there were no significant changes in the amounts of C3 and iC3b in PS19 tauopathy model mice

Figure 5. A 41-amino acid peptide of TREM2 alleviates microglial engulfment of synapses and ameliorates synaptic plasticity in 5xFAD mice

(A) The binding of C1q to synthetic TREM2 peptides encompassing residues 31–71 (41mer), 31–41, and 41–81 were assessed by SPB assay. The scrambled 41mer peptide was used as a control (three independent experiments; n = 6).

(B) The 41mer TREM2 peptide (10 μM) or C1-INH was incubated with C2 in the presence of C1 supplemented with or without fAβ42. C2 cleavage was analyzed by western blotting.

(C) Quantitation of the protein amounts of (C2a + C2b)/C2 in (B) (six independent experiments; n = 6 per group, two-way ANOVA).

(D) Increasing amounts of TREM2 41mer peptide or scrambled peptide control in NHS were added to IgM-coated microtiter plates, then C3 deposition was analyzed (four independent experiments; n = 8 per group, two-way ANOVA).

(E) WT or 5xFAD mice at 7-month-old age were injected with the scrambled peptide into the left hippocampus and the 41mer peptide into the right hippocampus. C3, iC3b, and GAPDH proteins were analyzed by western blotting 7 days after peptide injection.

(F) Quantitation of C3 and iC3b proteins in (E) (five independent experiments; n = 7 mice per group, two-way ANOVA).

(G) Coronal sections from peptide-injected 5xFAD mice were stained with PSD95 (green) and C3 (red); representative z-stack images of the hippocampal regions are shown. Scale bars, 10 μm.

(H–J) Quantitation of the density of C3 deposits (H), PSD95 puncta (J), and the percentage of C3 colocalized with PSD95 (I). Data points represent the average of two technical repeats (one experiment; n = 6 mice per group, paired Student's t test).

(K) Three-dimensional reconstruction and surface rendering of microglia, CD68, and engulfed PSD95 puncta (detected within microglial CD68$^+$ structures) in the hippocampi of peptide-injected 5xFAD mice. Scale bars, 5 μm.

(L) Quantitation of PSD95 engulfment within CD68$^+$ microglial structures. Data points represent the average of two technical repeats (one experiment; n = 6 mice per group, paired Student's t test).

(M and N) Flow cytometry analysis (M) and quantitation (N) of synaptic material (PSD95$^+$) engulfed by microglia in the hippocampi of 5xFAD mice after peptide injection (two experiments; n = 7 mice per group, paired Student's t test).

(O) Representative traces of whole-cell patch-clamp miniature excitatory postsynaptic currents (mEPSCs) recordings from the hippocampi of peptide-injected 5xFAD mice.

(P) Average frequency and amplitude of mEPSCs from (O) (seven independent experiments; n = 7 mice per group, 20 cells of scrambled peptide and 23 cells of 41mer for analysis, unpaired Student's t test).

(Q) Cumulative probability distributions of mEPSC inter-event intervals and amplitude of peptide-injected 5xFAD mice.

(R) Representative traces of miniature inhibitory synaptic currents (mIPSCs) recordings from the hippocampi of peptide-injected 5xFAD mice.

(S) Average amplitude and frequency of mIPSCs from (R) (seven independent experiments; n = 7 mice per group, 15 cells of scrambled peptide and 21 cells of 41mer for analysis, unpaired Student's t test).

(T) Cumulative probability distributions of mIPSC inter-event intervals and amplitude of peptide-injected 5xFAD mice.

All data are represented as mean ± SEM. *$p < 0.05$; **$p < 0.01$; ***$p < 0.001$; ****$p < 0.0001$; ns, not significant.

See also Figures S3–S5.

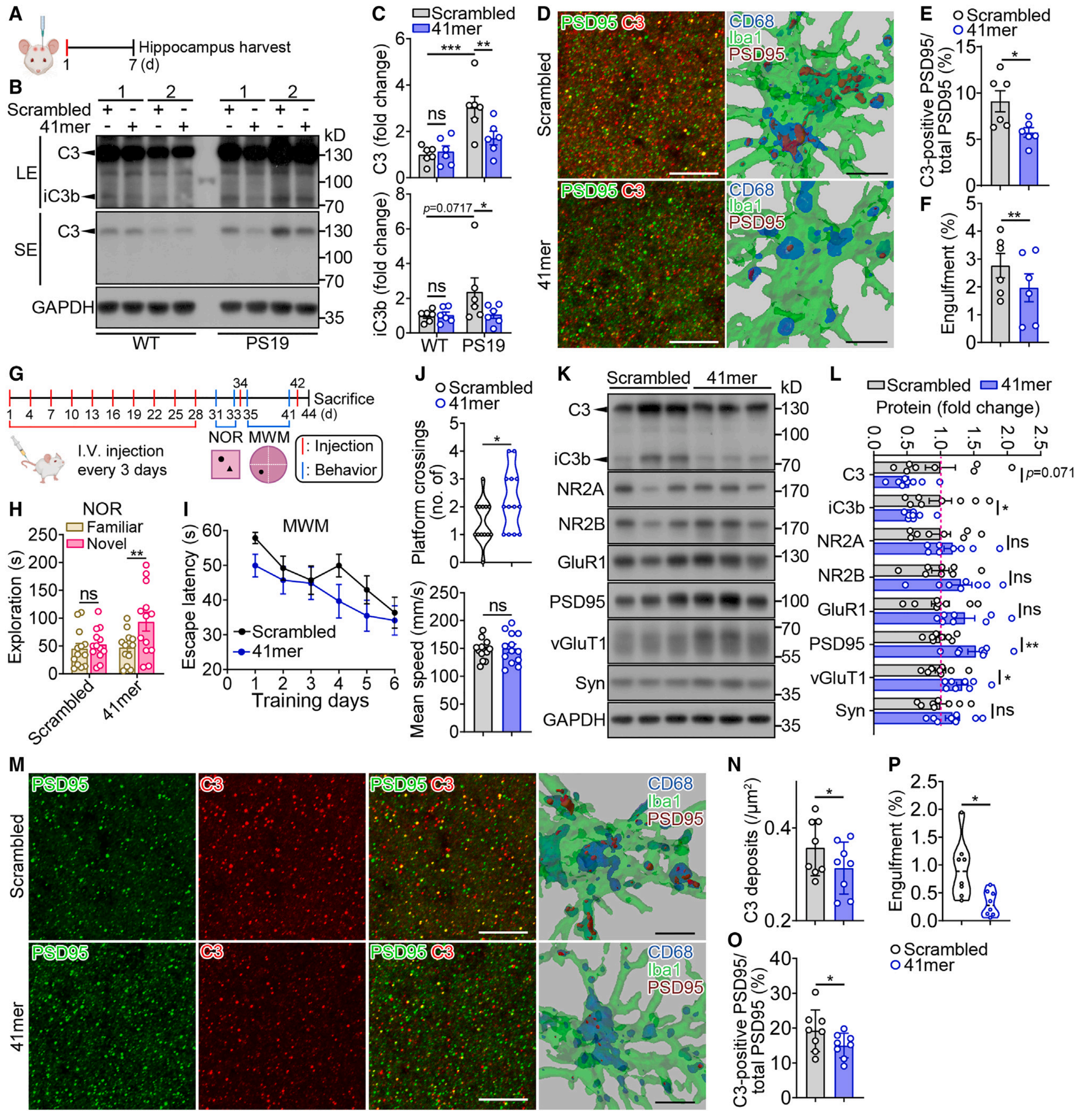

Figure 6. The TREM2 41mer peptide suppresses complement-mediated synaptic loss and enhances cognitive function in PS19 mice

(A) Cartoon illustrating the brain stereotaxic injection. WT or PS19 mice at 12-month-old age were injected with the scrambled peptide into the left hippocampi and the 41mer peptide into the right hippocampi for 7 days.

(B) Representative immunoblots of C3 and iC3b in hippocampal lysate from peptide-injected WT or PS19 mice.

(C) Quantitation of C3 and iC3b proteins in (B) (two independent experiments; n = 6 mice per group, two-way ANOVA).

(D) Representative confocal images of CA1 region co-stained with PSD95 and C3 in PS19 mice (scale bars, 10 μm), and the 3D rendering of confocal stacks of microglia and engulfed PSD95 puncta (scale bars, 5 μm).

(E) Quantitation of the percentage of C3 colocalized with PSD95. Data points represent the average of two technical repeats (one experiment; n = 6 mice per group, paired Student's t test).

(F) Quantitation of PSD95 engulfment within $CD68^+$ microglial structures. Data points represent the average of two technical repeats (one experiment; n = 6 mice per group, paired Student's t test).

(G) Schematic depiction of the experimental strategy. The 8-month-old PS19 mice were administered with the 41mer or the scrambled peptide via tail vein injection once every 3 days for 1 month.

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administered with the TREM2 41–81 peptide (Figures S3J and S3K). Consistent with the 7-month-old WT mice (Figures 5E and 5F), the 31–71 peptide had no significant impacts on the protein amounts of C3 and iC3b in 12-month-old WT mice (Figures 6B and 6C). A lower percentage of C3 colocalized with PSD95 and less PSD95 puncta in CD68^{+} microglial structures were observed upon TREM2 peptide administration in the PS19 mice, demonstrating that the TREM2 31–71 peptide restricts microglial engulfment of synapses (Figures 6D–6F). The tau pathology and microglial density assessed by AT8 and Iba1 staining, respectively, did not significantly differ between the TREM2 peptide and the scrambled peptide-control-treated groups (Figures S4G and S4H), suggesting a specific and direct impact of the TREM2 peptide on complement activity.

To validate the findings from the stereotaxic injection model and to further assess the long-term impact of the TREM2 peptide on complement activity, we administered the 8-month-old PS19 mice with a dose of 1.25 mg/kg TREM2 peptide once every three days through intravenous tail vein injection (Figure 6G). By using *in vivo* two-photon imaging, we found that the 41mer peptide rapidly entered the brain parenchyma after the intravenous injection without causing the blood-brain barrier leakage (Figure S6A). The novel object recognition test revealed that the 9-month-old PS19 mice treated with the scrambled peptide spent an equal amount of time exploring the novel and familiar objects, indicating an impairment of the recognition memory (Figure 6H). In contrast, the PS19 mice administered with TREM2 peptide spent significantly more time on the novel object than the familiar one, which implies a rescue of memory deficits. Treatment with the TREM2 peptide also rescued the spatial memory deficit of PS19 mice, as evidenced by increased platform crossing times in the hidden platform test of the Morris water maze (Figures 6I and 6J). Similar to the stereotaxic injection model, the TREM2 peptide significantly reduced the amounts of the iC3b protein and a trend toward decreased amounts of the C3 protein (Figures 6K and 6L). Immunostaining of brain sections further revealed a reduction of C3 deposition and a lower percentage of C3 colocalized with PSD95 (Figures 6M–6O). Decreased microglial engulfment of synaptic elements was correlated with the upregulation of synaptic proteins, including the presynaptic vGluT1 and the postsynaptic PSD95 (Figures 6K, 6L, and 6P). In contrast, injection of the 41mer peptide has minimal effects on complement activation, synaptic loss, or cognitive function in WT mice (Figures S6B–S6G). Although the TREM2 peptide suppresses complement-mediated synaptic loss and enhances cognitive function in PS19 mice, the tau pathology and microglial density were not significantly changed between the TREM2 peptide and the scrambled peptide-control-treated groups (Figures S4I and S4J). These data suggest that the TREM2 peptide protects against complement-mediated synaptic loss without significantly impacting tau pathology in PS19 mice.

DISCUSSION

Although it has been demonstrated that TREM2 is essential for microglia-mediated synaptic elimination in the developing brain, it remains unknown whether TREM2 regulates microglia-mediated synaptic pruning in the context of AD. Our current study unveils a role for TREM2 as a novel complement inhibitor via high-affinity binding to C1q through residues 31–71. The C1q-TREM2 complex prohibits complement-mediated synaptic elimination in the presence of AD pathology. Our work thus discovers a molecular mechanism through which TREM2 restricts complement-mediated synaptic elimination in the AD brain.

Among the AD-associated microglial genes, TREM2 has received much attention because the TREM2 R47H variant greatly increases the risk of developing AD. It has been widely acknowledged that TREM2 functions as a canonical receptor that initiates downstream signaling by interacting with a variety of ligands. Through unbiased screening, we report C1q as a new binding partner for TREM2 that binds to the extracellular domain of TREM2 with high affinity. However, unlike canonical ligands, which activate TREM2 signaling and modulate microglial functions in restricting AD pathology, TREM2 binding to C1q negatively regulates complement activity by sequestering C1q from tagging synapses for elimination by microglia. The density of TREM2-C1q complexes positively correlates with the amounts of synaptic proteins but negatively correlates with C3 deposition, suggesting a protective role of TREM2-C1q complex formation in the pathogenesis of AD. Both TREM2 and C1q proteins are predominantly expressed by microglia, which have previously been viewed as separately acting molecules performing distinct tasks in the brain. Our current data directly tie TREM2 to the regulation of the complement system in the brains of AD mouse models and AD patients.

Consistent with the previous study by Filipello et al.,[20] our current study also supports a role of TREM2 in neuronal circuit sculpting. Nevertheless, *Trem2* deficiency seems to yield

(H) Novel object recognition (NOR) task was used to assess recognition memory in PS19 mice after intravenous administration with the scrambled or 41mer peptide. Exploration time spent in the familiar and novel objects was measured for each mouse (one experiment; n = 13 mice per group; paired Student's t test).
(I and J) Spatial learning/memory was evaluated in the Morris water maze (MWM) in PS19 mice that received either the scrambled or 41mer peptide (one experiment; n = 13 mice per group; unpaired Student's t test). The escape latency to reach the hidden platform was recorded during the day 6 training (I). A probe trial on day 7 measured the platform crossing times and the swimming speed (J).
(K) Representative western blots of indicated proteins in hippocampal lysates from peptide-administered PS19 mice.
(L) Quantitation of C3, iC3b, and synaptic proteins in (K) (three independent experiments; n = 8 mice per group, unpaired Student's t test).
(M) Representative confocal images of PSD95 and C3 staining in the hippocampal CA1 region (scale bars, 10 μm) and the 3D rendering of confocal stacks of microglia and engulfed PSD95 puncta (scale bars, 5 μm).
(N and O) Quantitation of the density of C3 deposits (N), and the percentage of C3 colocalized with PSD95 (O). Data points represent the average of three technical repeats (one experiment; n = 8 mice per group, unpaired Student's t test).
(P) Quantitation of PSD95 engulfment within CD68^{+} microglial structures. Data points represent the average of two technical repeats (one experiment; n = 8 mice per group, unpaired Student's t test).
All data are represented as mean ± SEM. *p < 0.05; **p < 0.01; ***p < 0.001; ns, not significant.
See also Figures S4 and S6.

apparently different results on synaptic elimination during early development and adult-onset neurodegeneration. Lack of *Trem2* results in less synaptic elimination during neurodevelopment, whereas the deficiency of *Trem2* is paralleled by more synaptic elimination in neurodegeneration. This discrepancy is reminiscent of the well-documented differential roles of Trem2 depending on disease stage and model. Indeed, several studies have demonstrated that *Trem2* deficiency has disease-progression dependent effects on AD-related pathology and microglial cell functions.[41–44] Recognizing the dual roles of TREM2 on complement activity during early development and late neurodegeneration provides important insights into the complicated pathobiological functions of TREM2. Further investigation on the function and regulation of TREM2 signaling should be conducted to guide mechanism-based design of strategies for AD therapy by targeting TREM2.

Of note, complete deficiency of *Trem2* has been well documented to impair an array of microglial functions, including phagocytosis, migration, inflammation, survival, and proliferation.[16,45] Reduction in each of these activities can contribute to less microglial engulfment of synapse. Therefore, we chose to use the *Trem2* heterozygous mice in our current study, which may be a more relevant model to study how TREM2 contributes to AD pathogenesis. Consistent with a previous study,[35] *Trem2* haploinsufficiency significantly reduced microglial density and C1q protein amount in 5xFAD mice, thus complicating the data interpretation of the decreased amount of the C3 protein. In contrast, we observed negligible effects of *Trem2* haploinsufficiency on microglial density, C1q protein amount, or tau pathology in the early pathological stage of PS19 mice, allowing us to determine the specific and direct effects of *Trem2* deficiency on complement activation. The insufficiency of microglial *Trem2* decreases synaptic density by increasing complement-mediated microglial engulfment of synapses in the tauopathy mouse model, suggesting a critical role of TREM2 in preserving synaptic structure integrity.

Complement-mediated synaptic elimination plays an essential role in sculpting the circuitry of the nervous system and the fine-tuning of individual synaptic contacts. Emerging evidence suggests that the mechanisms of synaptic elimination become aberrantly reactivated, leading to pathological synaptic loss in neurodegenerative diseases.[1] In particular, C1q expression and C3 activation are markedly increased in AD mouse models and AD patients.[5,6] Inhibition of C1q and C3 rescues synaptic and cognitive deficits, suggesting that anti-complement therapeutic intervention could improve AD symptoms. In this study, we identified a 41-amino acid sequence in TREM2 that binds to C1q and inhibits C2 cleavage by the C1 complex, thereby reducing the downstream deposition of C3 *in vitro*. Furthermore, the synthetic TREM2 peptide reduces C3 deposition and synapse removal by microglia and rescues synapse density in both the amyloidosis and tauopathy mouse models of AD. It is interesting to note that the TREM2 peptide interferes with complement activation without altering microgliosis or the burden of Aβ and tau pathologies. The feature of this TREM2 peptide suggests that it can be a useful tool for further investigating the roles of TREM2 in the regulation of complement activity under human disease conditions.

The TREM2 ectodomain is cleaved by ADAM10/17 C-terminal to histidine 157, resulting in the release of soluble TREM2 (sTREM2) into the extracellular space.[46–49] Higher concentrations of sTREM2 in the cerebrospinal fluid (CSF) have been linked with slower disease progression in previous cross-sectional studies.[50–53] More recently, a longitudinal analysis shows that higher CSF concentrations of sTREM2 were correlated with a reduced rate of pathological progression in the presymptomatic phase of AD.[53] These findings together support a beneficial role of sTREM2 that protects against AD pathology. Because sTREM2 also possesses the C1q-binding region, it is tempting to speculate that sTREM2 might play a similar role to the full-length protein in sequestering C1q and diminishing the complement-mediated synaptic loss. Further studies are needed to evaluate the association between sTREM2 and C1q in human brains. Together, our findings add TREM2/sTREM2 to the repertoire of complement inhibitors in the brain and provide mechanistic insights into the protective roles of TREM2/sTREM2 against AD pathology.

Limitations of the study

Although we have employed multiple approaches to demonstrate a physical interaction between TREM2 and C1q, further studies are necessary to elucidate the underlying structural basis of this interaction. While our data indicate that the TREM2 R47H mutant exhibits a similar binding affinity to C1q compared with the wild-type TREM2 protein, structural investigations will provide valuable insights into the mechanisms by which the R47H variant does not impact TREM2's binding to C1q. Moreover, it is crucial to assess the effects of human TREM2 on complement binding and activity in humanized AD mouse models and larger human AD cohorts. Furthermore, it's important to repeat the TREM2 peptide injection experiments in AD/C1qa-KO mice when the mouse models are available in the future. The outcome of these experiments will add more evidence that the TREM2 peptide suppresses complement activation by interaction with C1q. Finally, because TREM2 is expressed on circulating monocytes and tissue-resident macrophages in the periphery, it is important to investigate whether peripheral Trem2 also plays a beneficial role in regulating complement activity. Addressing these questions will contribute to a more comprehensive understanding of the significance of TREM2-C1q interactions in the context of neurodegeneration.

STAR★METHODS

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SUPPLEMENTAL INFORMATION

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AUTHOR CONTRIBUTIONS

X.-F.C. and L. Zhong conceptualized and designed the project, analyzed and interpreted the data, and wrote the manuscript; L. Zhong, Y.L., and K.W. performed binding and complement activation experiments; L. Zhong, X.S., Y.L., and W.W. performed immunostaining and quantification on mouse brain tissues; X.S., L. Zhong, and D.-D.H. performed immunostaining and quantification on human brain tissues; X.S., Y.L., W.W., and L. Zhang did biochemical analyses; X.S. and Y.L. performed brain stereotaxic injection and tail vein injection; Y.H. and X.S. performed *in vivo* two-photon imaging; W.W. and H.R. performed mouse behavioral analysis; R.Z. and X.S. performed the electrophysiological experiment; W.W. performed the FACS experiment; Y.L., T.L., and Z.L. managed the mouse breeding and genotyping; L.C. performed part of the experiments in C2 cleavage and *E.coli* killing; M.C. did TREM2 41mer peptide binding experiment; G.B. participated discussion of the project and reviewed and edited the manuscript; X.-X.Y. provided material support; X.W. provided technical support. X.C. and Y.-w.Z. participated in discussion of the project and reviewed the paper. All the authors read and approved the final manuscript.

DECLARATION OF INTERESTS

X.-F.C., X.S., and L. Zhong are co-inventors on a patent application related to the work in this paper.

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
Rabbit polyclonal anti-NR2A	Millipore	Cat#07-632; RRID: AB_310837
Mouse monoclonal anti-NR2B	BD Biosciences	Cat#610416; RRID: AB_397796
Mouse monoclonal anti-GluR1	Millipore	Cat#MAB2263; RRID: AB_11212678
Rabbit monoclonal anti-PSD95	Cell Signaling Technology	Cat#3450S; RRID: AB_2292883
Mouse monoclonal anti-vGluT1	Millipore	Cat#MAB5502; RRID: AB_262185
Mouse monoclonal anti-Synaptophysin	Sigma-Aldrich	Cat#S5768; RRID:AB_477523
Mouse monoclonal anti-α-Tubulin	EMD Millipore	Cat#MABT205; RRID: AB_11204167
Mouse monoclonal anti-human Fc	R&D systems	Cat#MAB110; RRID: AB_884225
Rabbit polyclonal anti-GAPDH	Proteintech	Cat#10494-1-AP; RRID: AB_2263076
Mouse monoclonal anti-total tau (Tau5)	Thermo Fisher Scientific	Cat#AHB0042; RRID: AB_2536235
Rabbit monoclonal anti-Phospho-Tau (Thr181)	Thermo Fisher Scientific	Cat#701530; RRID: AB_2532491
Rabbit polyclonal anti-Phospho-Tau (Ser199/202)	Thermo Fisher Scientific	Cat#44-768G; RRID: AB_1502103
Goat polyclonal anti-TREM2 (Biotinylated)	R&D Systems	Cat#BAF1828; RRID: AB_2208688
Rabbit monoclonal anti-TREM2	Cell Signaling Technology	Cat#91068S; RRID: AB_2721119
Rabbit monoclonal anti-C3	Abcam	Cat#Ab200999; RRID: AB_2924273
Goat polyclonal anti-C3	MP biomedicals	Cat#0855730; RRID: AB_2334618
Mouse monoclonal anti-C1qa (Biotinylated)	Hycult Biotech	Cat#HM1096BT; RRID: AB_10258209
Goat polyclonal anti-C2	Complement Technology	Cat#A212
Rabbit polyclonal anti-C3	Santa Cruz	Cat#sc-20137; RRID: AB_634667
Goat polyclonal anti-C4	Complement Technology	Cat#A205; RRID: AB_2814828
Mouse monoclonal C5b-9	Santa Cruz	Cat#sc-58935; RRID: AB_1119839
Rat monoclonal anti-Mouse CD16/CD32	BD Biosciences	Cat#553141; RRID: AB_394656
Rat Monoclonal CD11b-APC	eBioscience	Cat#17-0112-82; RRID:AB_469343
Rat Monoclonal CD45-FITC	eBioscience	Cat#11-0451-85; RRID: AB_465051
Mouse monoclonal anti-PSD95	EDM Millipore	Cat#MAB1596; RRID: AB_2092365
Guinea pig polyclonal anti-PSD95	Synaptic Systems	Cat# 124 014 RRID: AB_2619800
Guinea pig polyclonal anti-vGluT1	EDM Millipore	Cat#AB5905; RRID: AB_2301751
Rabbit polyclonal anti-Iba1	Wako	Cat#019-19741; RRID: AB_839504
Rat monoclonal anti-CD68	BioLegend	Cat# 137001; RRID: AB_2044003
Mouse monoclonal anti-Aβ (MOAB-2)	Abcam	Cat#Ab126649

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
Goat polyclonal anti-ApoE	Santa Cruz	Cat#sc-6384; RRID:AB_634036
Rabbit polyclonal anti-App	Zhong et al.[40]	N/A
Mouse Monoclonal anti-Phospho-Tau (Ser202, Thr205)	Thermo Fisher Scientific	Cat#MN1020; RRID: AB_223647
Goat anti-Rabbit IgG (H+L) Secondary Antibody, HRP	Thermo Fisher Scientific	Cat#31460; RRID: AB_228341
Goat anti-Mouse IgG (H+L) Secondary Antibody, HRP	Thermo Fisher Scientific	Cat#31430; RRID: AB_228307
Rabbit anti-Goat IgG (H+L) Secondary Antibody, HRP	Thermo Fisher Scientific	Cat#31402; RRID: AB_228395
Bacterial and virus strains		
E. coli (DH5α transformed with pCDNA3.1-myc vector)	This paper	N/A
Biological samples		
Human brain tissue samples	Xiangya Human Brain Bank, Central South University	N/A
Chemicals, peptides, and recombinant proteins		
C1q	Complement Technology	Cat#A099
C1qa	OriGene	Cat#TP761199
C1qb	SinoBiological	Cat#10941-H08B
C1qc	OriGene	Cat#TP761200
C1	Complement Technology	Cat#A098
C2	Complement Technology	Cat#A112
C3	EMD Millipore	Cat#204885
ApoE3 protein	Fitzgerald	Cat#30R-2381
Streptavidin Poly-HRP40 Conjugate	Fitzgerald	Cat#65R-S104PHRP
human IgM	Calbiochem	Cat#401799
mannan	Sigma-Aldrich	Cat#M-7504
zymosan	Sigma-Aldrich	Cat#Z-4250
Aβ42	AnaSpec	Cat#AS-20276
BSA	Sigma-Aldrich	Cat#V900933
Normal human serum (NHS)	Complement Technology	N/A
C1q-depleted serum	Complement Technology	Cat#A300
Gelatin veronal buffer (GVB^{++})	Complement Technology	Cat#B100
GVBo buffer	Complement Technology	Cat#B103
C1-INH	Peprotech	Cat#130-20
TMB	Sigma Aldrich	Cat#T5569
ECL reagents	EMD Millipore	Cat#WBKLS0500
Protein A Agarose	Thermo Fisher Scientific	Cat#20334
Tetrodotoxin (TTX)	Tocris	Cat#1069
Percoll	Sigma-Aldrich	Cat#P1644
Dextran conjugated with Texas Red	Thermo Fisher Scientific	Cat#D1830
Scrambled peptide	Sangon Biotech	YLDWEHPGPVGKCDEVNDVQQSQSCE CSGMTALLWEHWLSD
TREM2 41mer peptide	Sangon Biotech	SLQVSCPYDSMKHWGRRKAWCRQLGEK GPCQRVVSTHNLWL
TREM2 31-41 peptide	Sangon Biotech	SLQVSCPYDSM
TREM2 41-81 peptide	Sangon Biotech	MKHWGRRKAWCRQLGEKGPCQRVVS THNLWLLSFLRRWNGS
C1qa 14-26 peptide	Sangon Biotech	AGRPGRRGRPGLK
TREM2 WT protein	This paper	N/A
TREM2 R47H protein	This paper	N/A
TREM2 D87N protein	This paper	N/A
TREM2-Fc protein	This paper	N/A
TREM2 41-81-Fc protein	This paper	N/A

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Continued		
REAGENT or RESOURCE	SOURCE	IDENTIFIER
Fc protein	This paper	N/A
TREM2 1-31-Fc protein	This paper	N/A
TREM2 1-51-Fc protein	This paper	N/A
TREM2 1-71-Fc protein	This paper	N/A
TREM2 1-91-Fc protein	This paper	N/A
TREM2 1-111-Fc protein	This paper	N/A
TREM2 1-131-Fc protein	This paper	N/A
TREM2 1-151-Fc protein	This paper	N/A
TREM2 1-171-Fc protein	This paper	N/A
Critical commercial assays		
Duolink® In Situ PLA® Probe Anti-Mouse PLUS	Sigma-Aldrich	Cat#DUO92001
Duolink® In Situ PLA® Probe Anti-Rabbit MINUS	Sigma-Aldrich	Cat#DUO92005
Duolink® In Situ Detection Reagents Green	Sigma-Aldrich	Cat#DUO92014
Duolink® In Situ Wash Buffers, Fluorescence	Sigma-Aldrich	Cat#DUO82049
Experimental models: Cell lines		
Primary neuronal cultures	This paper	N/A
Primary microglia	This paper	N/A
Experimental models: Organisms/strains		
Trem2$^{-/-}$	The Jackson Laboratory	JAX: 027197
5xFAD	The Jackson Laboratory	JAX: 034848
PS19	The Jackson Laboratory	JAX: 008169
Software and algorithms		
ImageJ	NIH	https://imagej.nih.gov/ij/
Imaris	Bitplane	http://www.bitplane.com/imaris/imaris
FlowJo	Tree Star	https://www.flowjo.com/
GraphPad Prism 8	GraphPad Software	https://www.graphpad.com
MiniAnalysis software	Synaptosoft	https://synaptosoft.software.informer.com/
TopScan Lite	Clever Sys. Inc	http://cleversysinc.com/?csi_products=topscan-lite
Deposited data		
Raw data and protocols for WB and IF with C3 and C1qa antibodies	This paper	http://dx.doi.org/10.17632/dvzvr3k9tw.1
Other		
Dounce homogenizer	Kimble Chase	Cat#KIM-885300-0002

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources and reagents should be directed to and will be fulfilled by the lead contact, Xiao-Fen Chen (chenxf@xmu.edu.cn).

Materials availability

Plasmids generated in this study are available upon request from the lead contact under a material transfer agreement.

Data and code availability

- All data reported in this paper will be shared by the lead contact upon request. The original data and detailed protocols for demonstrating the specificity of C3 and C1qa antibodies have been deposited in Mendeley Data: https://doi.org/10.17632/dvzvr3k9tw.1.

- This paper does not report original code.
- Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request.

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Mouse models

Trem2$^{-/-}$ (stock 027197), 5xFAD (stock 034848), and PS19 (stock 008169) mice were purchased from the Jackson Laboratory. The 5xFAD or PS19 mice were crossed to Trem2$^{-/-}$ mice to generate 5xFAD/Trem2$^{+/-}$ and PS19/*Trem2*$^{+/-}$. All mice were housed under a standard 12-hour light/dark cycle. Both males and females were used in this study. Sample sizes were adequately powered to observe the effects on the basis of past experience of animal studies.[40] Mice were randomly selected for further biological analysis, and the investigators were blinded to group allocation during the experiments or outcome assessments. All animal procedures were carried out under protocols approved by the Animal Ethics Committee of the Xiamen University.

For brain stereotaxic injection: The 5xFAD mice at 7 or 12 months of age and the PS19 mice at 12 months of age were injected with 10 μg TREM2 41mer peptide (SLQVSCPYDSMKHWGRRKAWCRQLGEKGPCQRVVSTHNLWL) or the scrambled peptide (YLDWEHPGPVGKCDEVNDVQQSQSCECSGMTALLWEHWLSD) into the right and left hippocampi, respectively. Seven days after injection, mice were anesthetized and perfused with ice-cold PBS. Hippocampi were harvested for biochemical or histological analysis.

For intravenous administration, 8-month-old PS19 mice were administrated with the TREM2 41mer peptide or the scrambled control via tail vein injection once every 3 days for one month (at a dose of 1.25 mg/kg). After behavioral tests, mice received one additional dose before being sacrificed. Brain tissues were collected for biochemical analysis or immunofluorescent staining.

Human samples

Post-mortem brain tissue (see Table S1) was obtained from patients with pathology-confirmed AD from Xiangya Human Brain Bank, Central South University. Human samples were procured with Ethics Committee approval and written informed consent.

METHOD DETAILS

Brain stereotaxic injection

Stereotaxic injection was performed as described previously.[40] Briefly, mice were anesthetized by isoflurane in a cage and placed in a stereotaxic frame (RWD Life Science, China). The anesthesia was maintained using isoflurane (1-2%) inhalation, then a skin incision was made, and holes were drilled at x (± 2.0 mm from bregma) and y (− 2.0 mm from bregma). Peptide solution was delivered to hippocampus at z-depths of 2.0 mm with 0.20 μL/min. After injection, the syringe was left in site for 5 min before being withdrawn slowly.

Novel object recognition

The novel object recognition test was performed in an open field apparatus (40 × 40 cm). On day one, mouse was placed in the open field arena and allowed to explore freely for 10 min. On day two (training session), two identical objects were placed symmetrically in opposite corners of the chamber, and mouse was allowed to explore the two objects for 10 min. Between trials, the objects and the chamber were cleaned with 75% ethanol. On day three (test session), one of the training objects was replaced with a novel object. Mouse was allowed to explore for 10 min, the time that mouse spent on exploring the novel object was used to measure recognition memory.

Morris water maze

Experiments were conducted in a circular 120-cm-diameter pool filled with opaque water kept between 20 and 22°C. The mice were given two training trials per day for six consecutive days to find a hidden platform that submerged 1 cm below the water surface. The time taken to reach the platform (escape latency) was measured, and the average of two trials was determined. In each trial, the searching time for each mouse was no more than 60 s. Each mouse remained on the platform for 15 s if they found it or they were guided to the platform and stayed for 10 s. A probe trial without a platform was performed 24 h after the last trial of the hidden platform test. Mice were tested to find the platform from the opposite quadrant for 60 s. The crossing times to the platform regions and the swimming velocity were recorded. Mice were tracked by a video camera (SONY) in both training and probe trials. Collected data were analyzed by TopScan Lite software (Clever Sys. Inc).

Tissue preparation

Mice were anesthetized by isoflurane and transcardially perfused with 50 mL ice-cold PBS. Subsequently, brains were removed from the skulls. For immunofluorescent staining, brains were fixed in 4% paraformaldehyde (PFA, Sigma-Aldrich, P6148) overnight at 4°C, then immersed in 30% sucrose solution for at least 48 h for cryoprotection. The brains were embedded in Tissue-Tek® O.C.T. Compound (SAKURA, 4583), and subsequently mounted on a freezing microtome (Leica CM 1950) to obtain 15-μm thick coronal sections. For biochemical analysis, the cortex and hippocampus were dissociated and immediately snap-frozen in liquid nitrogen and stored at −80°C for extraction of protein and RNA.

Immunofluorescent staining

For brain sections, fifteen-micron cryosections were mounted on poly-L-lysine precoated glass slides, air-dried overnight at 37°C, and subsequently processed for staining. Sections were incubated in sodium citrate solution (10 mM) at 100°C for antigen retrieval for 15min. Once cooled, sections were washed three times in PBS and then blocked at room temperature (RT) for 1 h in blocking buffer (PBS with 5% normal donkey serum and 0.5% Triton X-100). Slices were incubated with primary antibodies overnight at 4°C, washed three times, then incubated with Alexa-fluorophore-conjugated secondary antibodies (Thermo Fisher Scientific, A-21202, A-21207, or A-21450, 1:400) for 2 h at RT in the dark. Sections were washed and sealed with an anti-fade reagent (Life Technologies, P36935). The following antibodies for immunofluorescent detection of brain sections were used in this study: mouse anti-PSD95 (EDM Millipore, MAB1596, 1:100), rabbit anti-C3 (Abcam, ab200999, 1:100), guinea pig anti-vGluT1 (EDM Millipore, AB5905, 1:100), rabbit anti-Iba1 (Wako, 019-19741, 1:200), rat anti-CD68 (BioLegend, 137001, 1:100), mouse anti-Aβ (MOAB-2, Abcam, ab126649, 1:500), mouse anti-Phospho-Tau (Ser202, Thr205) (AT8, Thermo Fisher Scientific, MN1020, 1:400).

For primary neuronal cultures, cells were washed three times in PBS and fixed with 4% PFA for 30 min at RT. After rinsing with PBS, neurons were permeabilized with 0.2% Triton X-100 in PBS for 5 min and blocked with 5% BSA in PBS for 1 h at RT. Cells were incubated with rabbit anti-PSD95 (CST, 3450S, 1:100) and mouse anti-Synaptophysin (Sigma-Aldrich, S5768, 1:200) primary antibodies overnight at 4°C. Cells were washed three times and incubated with Alexa-fluorophore-conjugated secondary antibodies for 2 h at RT. Finally, cells were washed and sealed with an anti-fade reagent (Life Technologies, P36935).

Proximity ligation assay

Proximity ligation assay (PLA) was performed with the Duolink PLA mouse/rabbit kit (Sigma-Aldrich) according to the manufacturer's instructions. Briefly, fixed postmortem human hippocampal sections were washed with PBS and processed for antigen retrieval, then permeabilized in 0.5% Triton X-100 (in PBS) for 30 min and blocked with blocking solution at 37°C for 1 h. Rabbit anti-TREM2 (CST, 91068S, 1:100), mouse anti-C1qa (Hycult Biotech, HM1096BT, 1:100), goat anti-C3 (MP biomedicals, 0855730, 1:50), guinea pig anti-PSD95 (Synaptic Systems, 124014, 1:100) or guinea pig anti-vGluT1 (EDM Millipore, AB5905, 1:100) antibodies were added and incubated at 4°C overnight. PLA probe anti-mouse PLUS (Sigma-Aldrich, DUO92001) and PLA probe anti-rabbit MINUS (Sigma-Aldrich, DUO92005) secondary antibodies were incubated at 37°C for 1 h. After washing, sections were subjected to ligation at 37 °C for 30 min, followed by amplification with polymerase for 100 min at 37 °C. Finally, brain sections were washed with buffer B (Sigma-Aldrich, DUO82049-4L) three times. For C3, PSD95 or vGLUT1 staining, brain sections were incubated for 2 h with Alexa Fluor™ 647-conjugated donkey anti-goat (Invitrogen, A21447, 1:200) or goat anti-guinea pig (Invitrogen, A21450, 1:200) secondary antibodies, washed, and sealed with an anti-fade reagent (Life Technologies, P36935). PLA without two primary antibodies was used as a technical control.

Microscopy and analysis

For confocal microscopy, images were scanned and acquired with a NIKON A1R Plus confocal microscope. Identical imaging parameters were maintained constant in the same experiment. For C3 deposition on synapse, 4-μm z-stacks consisting of 5 optical slices of 1-μm thickness were captured on a 60x objective with 3x digital zoom. The density of C3 and synaptic proteins were analyzed using the ImageJ. For microglial engulfment, 8-μm z-stacks consisting of 9 optical slices of 1-μm thickness were captured on a 60x objective with a 3x zoom, then Imaris software (Bitplane) was used to perform 3D reconstruction and surface rendering, as described previously.[54]

Neuron-microglia co-cultures and peptide treatment

Primary neuronal and microglial cultures were prepared as described in previous studies.[55,56] After centrifugation at 800 g for 5 min, microglia were resuspended in Neurobasal medium and plated onto neurons at DIV 11 (1:10 ratio). For peptide treatment, the scrambled control or the TREM2 41mer peptide was added to treat cells for 24 h.

Western blotting

Brain tissues were homogenized in RIPA Lysis and Extraction Buffer (Thermo Fisher Scientific, 89900) supplemented with Protease and Phosphatase Inhibitor Cocktail (Thermo Fisher Scientific, 78440). The total protein concentration of each sample was determined by BCA Assay Kit (Thermo Fisher Scientific, 23225), and equal amounts of total protein were loaded onto 4-12% Bis-Tris NuPAGE gels (Thermo Fisher Scientific, NP0322BOX). Following electrophoresis, proteins were transferred to a 0.45 μm PVDF membrane (EMD Millipore, IPVH00010). After blocking, blots were probed overnight at 4°C with primary antibodies and next incubated with horseradish peroxidase (HRP)-conjugated secondary antibodies (Thermo Fisher Scientific, 31460, 31430, 31402, 1:2000) for 2 h at RT. ECL reagents (EMD Millipore, WBKLS0500) were used for detection of the signal. Immunoreactive bands were quantified using ImageJ software. The following antibodies were used: rabbit anti-C3 (Abcam, ab200999, 1:1000), rabbit anti-TREM2 (CST, 91068S, 1:1000), mouse anti-C1qa (Hycult Biotech, HM1096BT, 1:1000), goat anti-C2 (Complement Technology, A212, 1:1000), rabbit anti-NR2A (Millipore, 07-632, 1:2000), mouse anti-NR2B (DB Biosciences, 610416, 1:2000), mouse anti-GluR1 (Millipore, MAB2263, 1:2000), rabbit anti-PSD95 (Cell Signaling Technology, 3450S, 1:2000), mouse anti-vGluT1 (Millipore, MAB5502, 1:2000), mouse anti-Synaptophysin (Sigma-Aldrich, S5768, 1:2000), mouse anti-α-Tubulin (EMD Millipore, MABT205, 1:5000), mouse anti-human Fc (R&D systems, MAB110, 1:1000), rabbit anti-GAPDH (Proteintech, 10494-1-AP, 1:50000), mouse anti-total tau (Tau5, Thermo Fisher Scientific, AHB0042, 1:1000), rabbit anti-Phospho-Tau (Thr181, Thermo Fisher Scientific, 701530, 1:1000), rabbit anti-Phospho-Tau (Ser199/202, Thermo Fisher Scientific, 44-768G, 1:1000).

RNA isolation and RT-qPCR

Total RNA was extracted from primary microglia or brain lysate using TRIzol reagent (Thermo Fisher Scientific, 15596018). One microgram of RNA was used in 20 μL of reverse transcription reaction (TOYOBO, FSQ-201). Quantitative PCR was performed using the FastStart Universal SYBR Green Master mix (Roche, 04913914001) on 7500 fast (ABI) or 480 LightCycler (Roche). The $\Delta\Delta$Ct method was applied to determine differences in gene expression after normalization to the arithmetic mean of β-Actin. The primer sequences were as follows: β-Actin-Forward: AGTGTGACGTTGACATCCGTA, β-Actin-Reverse: GCCAGAGCAGTAATCTCCTTC; IL1β-Forward: GCAACTGTTCCTGAACTCAACT, IL1β-Reverse: ATCTTTTGGGGTCCGTCAACT; IL6-Forward: CAATGGCAATTCTG ATTGTATG, IL6-Reverse: AGGACTCTGGCTTTGTCTTTC; TNF-Forward: CCCTCACACTCAGATCATCTTCT, TNF-Reverse: GCTAC GACGTGGGCTACAG.

Affinity purification-mass spectrometry (AP-MS)

The brains of Trem2$^{-/-}$ mice were homogenized in lysis buffer (1% Nonidet P-40, 50 mM Tris-HCl, pH 8.0, 150 mM NaCl) supplemented with protease inhibitors. Homogenates were centrifuged at 12,000 g for 30 min. Supernatants were collected and protein A agarose beads (Thermo Fisher Scientific, 20334) pre-coupled to purified human Fc control or recombinant TREM2-Fc protein were added and incubated overnight at 4°C. The agarose resin with bound proteins was washed three times with lysis buffer. Bound proteins were eluted from the beads by incubation with elution buffer (Thermo Fisher Scientific, 21004) and precipitated with TCA overnight at 4°C, followed by centrifugation at 16,000 g for 10 min. Protein pellets were washed with ice-cold acetone and dried briefly by vacuum centrifugation, then protein precipitates were resuspended in 8 M urea. The samples were subsequently reduced by dithiothreitol (5 mM, 30 min), and alkylated in the dark by iodoacetamide (15 mM, 30 min). Proteins were digested with trypsin overnight at 37°C and the reaction was stopped by addition of formic acid to 1% final concentration. Peptides were loaded onto a StageTip for desaltion, and eluted directly into inserts with 70% acetonitrile/1% formic acid. The tryptic peptides were then used for MS analysis, which was performed on an AB Sciex TripleTOF 5600 mass spectrometer interfaced to an Eksigent NanoLC Ultra 2D Plus HPLC system.

Solid-phase binding

Solid-phase binding was performed in a 96-well plate using enzyme-linked immunosorbent assay (ELISA). C1q (Complement Technology, A099), C1qa (OriGene, TP761199), C1qb (SinoBiological, China, 10941-H08B), C1qc (OriGene, TP761200), C1 (Complement Technology, A098), C2 (Complement Technology, A112), or C3 (EMD Millipore, 204885) in PBS was immobilized onto plate wells overnight with 100 μL per well. After the plates were washed three times with 0.1% PBST (PBS with 0.1% Tween 20) and blocked with 4% BSA for 1 h at 37°C, purified TREM2 protein (homemade,[55] 0-50 nM) diluted in PBS was added and incubated for 1 h at 37°C. After washing, the bound proteins were detected with TREM2 biotinylated antibody (R&D Systems, BAF1828, 1:2000) for 1 h at 37°C. After primary antibody incubation, plates were washed and then incubated with Streptavidin Poly-HRP40 Conjugate (Fitzgerald, 65R-S104PHRP, 1:3000) for 30 min at 37°C. Finally, plates were washed again and developed with TMB (Sigma Aldrich, T5569) and the absorbance at 656 nm was recorded. Alternatively, TREM2 protein (100 nM), apoE3 (Fitzgerald, 30R-2381, 100 nM), TREM2 41mer or scrambled peptide (Sangon Biotech, China, 10 μM) was immobilized on the plates, while C1q proteins diluted in PBS (0-50 nM) were subsequently incubated to bind each target. The bound proteins were detected by using a biotinylated-anti-C1qa antibody (Hycult Biotech, HM1096BT, 1:2000).

Surface plasmon resonance

Surface plasmon resonance (SPR) analysis was performed at 25°C using a Biacore T200 (GE Healthcare). Purified TREM2 protein was immobilized onto Biacore CM5 chip (GE Healthcare, BR-1005-30) using the amine coupling kit (GE Healthcare, BR-1000-50) following the manufacturer's instructions. C1q was tested with a gradient concentration of 15.63, 31.25, 62.50, 125.0, and 250.0 nM, respectively, and the response was measured in resonance units. Data were analyzed using Biacore T100 Evaluation software.

Dot blot

C1q, C1qa, C3, and apoE3 proteins diluted in PBS were spotted onto a PVDF membrane using a dot blot manifold apparatus (GE Healthcare). Membrane strips were blocked with 5% milk in PBS for 2 h at RT, and incubated with TREM2 protein (2 ng/mL in PBS) overnight at 4°C. After washing three times with 0.1% PBST, bound proteins were detected with human TREM2 biotinylated antibody (R&D Systems, BAF1828, 1:2000) and followed by incubation with Streptavidin Poly-HRP40 Conjugate (Fitzgerald, 65R-S104PHRP, 1:3000). Finally, the signal was detected using ECL reagents (EMD Millipore, WBKLS0500).

Pull-down assay

For the Fc pull-down assay, purified human Fc and TREM2-Fc were immobilized on protein A agarose beads (Thermo Fisher Scientific, 20334). Beads were then incubated with brain lysates from Trem2$^{-/-}$ mice overnight at 4°C in lysis buffer (1% Nonidet P-40, 50 mM Tris-HCl, pH 8.0, 150 mM NaCl). After three washes in lysis buffer, the bound C1qa was detected and analyzed by Western blotting. For the biotin pull-down assay, $Trem2^{-/-}$ mouse brain lysates were incubated with biotin-labeled scrambled or 41mer peptide at 4°C overnight. Streptavidin agarose beads (Thermo Fisher Scientific, 20353) were added and incubated at 4°C for 2 h. After washing three times, C1qa, App and apoE were detected by Western blotting.

Complement activation assay

Complement activation was assessed by C3b, C4b, or C5b-9 (TCC) deposition using ELISA method. Microtiter plates (Nunc) were incubated overnight at 4°C with human IgM (Calbiochem, 4017991MG, 2 μg/mL), mannan (Sigma-Aldrich, M-7504, 100 μg/mL), zymosan (Sigma-Aldrich, Z-4250, 100 μg/mL) or Aβ42 fibrils (AnaSpec, AS-20276, 5 μM) in 75 mM sodium carbonate (pH 9.6). Human IgM, mannan, and zymosan specifically activate the classical pathway (CP), lectin pathway (LP), and alternative pathway (AP), respectively. Between each step of the procedure, the plates were washed three times with 0.1% PBST. The wells were blocked with 2% BSA (Sigma-Aldrich, V900933) in PBS for 2 h at RT. For CP or LP, normal human serum (NHS) (Complement Technology) was diluted in gelatin veronal buffer (GVB^{++}, Complement Technology, B100) and used at a concentration of 1% for measurement of CP, 5% for LP, 5% for fAβ42-induced complement activation. In the case of AP, C1q-depleted serum (Complement Technology, A300) was diluted in GVBo buffer (without Ca^{2+} and Mg^{2+}, Complement Technology, B103) supplemented with Mg^{2+}-EGTA (10 mM) at a concentration of 20%. To test the effects of TREM2 or TREM2 41mer peptide on complement activation, the diluted serum was preincubated with various concentrations of TREM2 or the 41mer peptide for 2 h at 4°C, then added to the plates for 25 min at 37°C. Complement activation was assessed by detecting deposited complement factors using specific antibodies against C3 (Santa Cruz, sc-20137, 1:100), C4 (Complement Technology, A205, 1:3000), and C5b-9 (Santa Cruz, sc-58935, 1:100) in the blocking buffer for 1 h at RT. Plates were then incubated with HRP-labeled anti-rabbit, anti-goat, or anti-mouse secondary antibodies (Thermo Fisher Scientific, 31460, 31402, 31430, 1:2000) for 30 min at RT. Finally, plates were developed with TMB (Sigma Aldrich, T5569) and the absorbance at 656 nm was recorded.

C2 cleavage assay

The effects of TREM2 and TREM2 41mer peptide on C2 cleavage by C1 were determined in fluid phase. Briefly, TREM2 recombinant protein (4 μM), the 41mer peptide (10 μM) or C1 esterase inhibitor (C1-INH) (Peprotech, 130-20, 5 μg/mL) was preincubated with C1 (0.1 μg/mL) in GVB^{++} buffer for 2 h at 4 °C. C2 (2 μg/mL) supplemented with or without fAβ42 (2 μM) was added and incubated at 37°C for 1h. The C1-INH was used as a positive control. Finally, the cleavage reaction was stopped by addition of equal amounts of 2x SDS loading buffer, and samples were boiled for 10 min. C2 cleavage was analyzed by Western blotting.

E. coli killing assay

The assay was performed as described previously.14 Briefly, Fc or TREM2-Fc was preincubated with 1 % NHS (in GVB^{++} buffer) for 2 h at 4 °C, followed by adding *E. coli* (DH5α transformed with pCDNA3.1-myc vector) and incubating for 30 min at 37°C. Colony-forming units were counted after plating on an LB-agar plate and incubation at 37°C overnight. Bacterial survival without NHS (0 % NHS) was set as 100%.

Synaptic phagocytosis assay

The 7-month-old 5xFAD mice were injected with TREM2 41mer peptide or the scrambled peptide into the hippocampi as described above. After 7 days, mice were anesthetized and transcardially perfused using ice-cold PBS. The hippocampi were quickly dissected and chopped into pieces, then homogenized with a Dounce homogenizer (Kimble Chase, KIM-885300-0002). The homogenates were filtered through a pre-wet 100-μm cell strainer, centrifuged at 800 g for 5 min at 4 °C. To remove debris and myelin, cells were resuspended in 10 mL of 30% Percoll (Sigma-Aldrich, P1644) and spun down for 40 min at 800 g. The cell pellet was washed with 10 mL of ice-cold PBS and spun again for 5 min at 800 g. Cells were incubated with the Fc receptor blocking antibody CD16/CD32 (BD Bioscience, 553141, 1:100) for 10 min at 4 °C to prevent unspecific binding, followed by 15 min incubation with CD11b-APC (eBioscience, 17-0112-82, 1:100) and CD45-FITC (eBioscience, 11-0451-85, 1:100). For intracellular staining, cells were fixed and permeabilized (BD Biosciences, 554714) before staining with rabbit anti-PSD95 (Cell Signaling Technology, 3450S, 1:200), then incubated with a secondary fluorescent antibody (Thermo Fisher Scientific, A10042, 1:200) for 15 min. Finally, cells were washed again and resuspended in 2% FBS/PBS for FACS analysis on a CytoFlex LX (Beckman). The data were analyzed using FlowJo (Tree Star).

Electrophysiology

Whole-cell voltage-clamp recordings were carried out as previously reported.[40] For brain tissue, mice were anesthetized with isoflurane, and the brain was quickly removed and dissected in an ice-cold oxygenated (95% O_2/5% CO_2) cutting solution containing 64 mM NaCl, 2.5 mM KCl, 1.25 mM NaH_2PO_4, 10 mM $MgSO_4$, 0.5 mM $CaCl_2$, 26 mM $NaHCO_3$, 10 mM glucose, and 120 mM sucrose. Acute coronal slices (400 μm) prepared using a Vibroslice (VT 1200 S; Leica) were recovered for 30 min at 32 °C and then at RT for at least 1 h before recording in the artificial CSF (aCSF) containing 126 mM NaCl, 2.5 mM KCl, 1.2 mM NaH_2PO_4, 2.4 mM $MgCl \cdot 6H_2O$, 1.2 mM $CaCl_2$, 18 mM $NaHCO_3$, and 11 mM Glucose. Slices were transferred to the recording chamber and superfused with aCSF (2 mL/min) saturated with 95% O_2/5% CO_2 at 32 °C. The patch pipette with a resistance of 5–9 MΩ was filled with solution containing 140 mM $CsCH_3SO_3$, 2 mM $MgCl \cdot 6H_2O$, 5 mM TEA-Cl, 10 mM HEPES, 2.5 mM MgATP, 0.3 mM Na_2GTP, and 1 mM EGTA (pH 7.3 adjusted with CsOH, 290 mOsm). Pyramidal neurons in CA1 were visualized with infrared optics using an upright fixed microscope equipped with a 40X water-immersion lens (FN1, Nikon) and CCD monochrome video camera (IR1000, DAGE-MTI). mEPSC and mIPSC were recorded in the presence of 1 μM tetrodotoxin (TTX; Tocris, Cat. No. 1069) in voltage-clamp mode at a holding potential of −65 mV and 0 mV, respectively. Series resistance was monitored throughout the experiments and was below 30 MΩ. Whole-cell recording was performed with MultiClamp 700B (Molecular Devices) amplifier and 1550 digitizer (Molecular Devices). Data were filtered at 2 kHz, sampled at 10 kHz with pClamp software (Molecular Devices, version 10.6), and analyzed off-line with MiniAnalysis software (Synaptosoft Inc., Fort Lee, NJ). For primary neuronal cultures at DIV 12-14, recordings were performed as described above.

In vivo two-photon imaging

The FITC-labeled peptide in the brain was imaged by two-photon microscopy. Briefly, PS19 mice were anesthetized by the inhalation of isoflurane (1.5 % for induction, 1.0 % for surgery, and 0.75% for imaging) and immobilized on a stereotactic apparatus. For imaging under the cranial window, a 4-mm-diameter craniotomy over the cerebral cortex was made, and a 4-mm coverslip was added onto the exposed cortex. During imaging, dextran conjugated with Texas Red (Thermo Fisher Scientific, D1830) and the FITC-labeled peptide was administered by intravenous injection. Mice were imaged using a two-photon imaging system (FVMPE-RS, Olympus) equipped with a dual-line femtosecond laser tunable to 680–1300 nm and a fixed 1040 nm laser (Spectra Physics). A ×25 water immersion lens (NA, WD, Olympus) was used for fluorescence imaging *in vivo*, and a 512 × 512 pixels field was viewed. Laser wavelength was tuned to 830 nm to excite the fluorescence of FITC and Texas Red, and emission light was detected with 495/540 and 575/645 filters, respectively.

QUANTIFICATION AND STATISTICAL ANALYSIS

Statistical tests were performed using GraphPad Prism 8 (GraphPad Software, La Jolla, CA). All the immunofluorescent and electrophysiological data analyses were performed blindly. For complement deposition experiments, statistical significance was determined using unpaired Student's t test or one-way ANOVA. Paired Student's t test or unpaired Student's t test was used for data analysis in brain stereotaxic injection experiments. Two-way ANOVA followed by Tukey post hoc analyses was used to analyze C2 cleavage. Detailed information on the statistical method for each quantitated data was described in each figure legend. All data were shown as mean ± SEM and specific n values were provided in each figure legend. Levels of significance are as follows: $^*p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$, $^{****}p < 0.0001$. $p < 0.05$ was considered as statistically significant.